

How AapdaNetra Generates Risk Scores: Jury Pitch Script

Verbatim 45-Second Defense Script, Formula Deconstruction & Whiteboard Map

1. The 45-Second Spoken Pitch (Word-for-Word Script for Judges)

■■ VERBATIM PRESENTATION SCRIPT (Memorize & Deliver with Confidence)

“Respected Jury, a fundamental flaw in existing disaster systems is that they rely either on **raw rainfall data** (which completely ignores local topography) or **opaque black-box AI** (which government officials cannot defend in court).

In **AapdaNetra**, we generate our **0–100 Unified Risk Score** using an explainable, 5-pillar composite index:

- 1. 40% Predictive AI:** High-precision XGBoost classifier trained on real river basin hydrology with **96.2% recall**.
- 2. 25% Real-Time Physics:** Live rainfall, humidity, and wind sensitivity curves from OpenWeather.
- 3. 15% Historical GIS Memory:** Geospatial 15km scanning of past disaster polygons in MongoDB.
- 4. 10% Infrastructure Vulnerability:** Mean structural fragility of habitations within 10km.
- 5. 10% Citizen Ground Truth:** Real-time verified crowdsourced reports from citizens on the ground.

If the score crosses **76**, it enters **CRITICAL** status, triggering our smart acoustic sentinel and mandatory evacuation protocols. And if external cloud AI servers drop offline during a storm, our engine **instantly falls back to deterministic physics rules** with zero downtime.”

2. The 5-Pillar Mathematical Equation & Weight Breakdown

Pillar Component	Weight	Underlying Technology / Dataset	Strategic Justification for Judges
1. Predictive ML	40%	XGBoost / Random Forest on 6 features (Rain, Water Level, Elevation, Humidity, History, Temp).	Captures non-linear relationships that simple rainfall rules miss.
2. Real-time Physics	25%	Live OpenWeather telemetry mapped to hazard sensitivity curves.	Immediate reaction to cloudbursts without waiting for batch runs.
3. Historical GIS Memory	15%	MongoDB 2dsphere proximity query (15km radius) on HazardZone polygons.	Low-lying historic floodplains are inherently higher risk.
4. Infrastructure Vulnerability	10%	Mean socio-economic & housing fragility index within 10km.	Kuccha houses and slums face higher threat than concrete towers.
5. Citizen Ground Truth	10%	Verified citizen reports in last 24h: min(Reports × 5, 20).	Direct crowdsourced ground validation of blocked drains & floods.

3. Action Tiers & Automated System Response

Score Band	Threat Tier	AapdaNetra Automated System Action	Standard Operating Procedure (SOP)
76 – 100	CRITICAL	Smart Acoustic Siren Sounds (if user in zone) + Brevo REST Emergency Broadcast.	MANDATORY EVACUATION: Move low-lying settlements to verified shelters.
51 – 75	RED	Warning banner on dashboard; shelters placed on standby intake.	PREPARE EVACUATION: Pre-position rescue boats and emergency rations.
26 – 50	AMBER	Advisory toast popup; sluice gate monitors placed on alert.	WATCH & MONITOR: High-frequency sensor polling every 5 minutes.
0 – 25	GREEN	Normal status; 100% silent audio; baseline telemetry.	ROUTINE OPS: Normal administrative monitoring.

4. The 3 'Judge-Killer' Defense Counter-Arguments

When an evaluator attempts to challenge your formula, deliver these exact counter-points:

- Challenge: 'Why not just use an end-to-end Deep Neural Network or pure AI?'
Winning Counter: 'A pure deep neural network is a black box. An IAS District Magistrate cannot order the evacuation of 50,000 residents without knowing the exact causal drivers. Our formula provides Explainable AI (XAI) feature weight decomposition: the magistrate sees that 40% comes from dam discharge and 25% from local cloudbursts, making the decision legally defensible.'
- Challenge: 'Why not just use rainfall thresholds like the meteorological department?'
Winning Counter: 'Rainfall alone is dangerously misleading. 50mm of rain on dry plains causes puddles, while the exact same 50mm on a deforested slope with saturated soil causes a devastating landslide. Our formula incorporates terrain elevation, antecedent soil moisture, and historical inundation zones.'
- Challenge: 'What happens if the AI server loses internet during a cyclone?'
Winning Counter: 'We designed for disaster resilience. If our Python microservice drops connection, our Node.js backend automatically falls back to deterministic rule-based physics equations using spatial hashing and soil saturation. The life-safety alert engine never halts.'

5. Whiteboard Diagram Blueprint (Draw this in 10 Seconds)

Draw this exact 1-line formula on the whiteboard:

[Risk Score (0-100)] = [40% ML Probability] + [25% Live Weather Physics] + [15% Historic GIS Zones] + [10% Habitation Fragility] + [10% Citizen Reports]

Then draw an arrow pointing to 76+:

■■■■■ If Score ≥ 76 → [CRITICAL] → [Triggers Smart Acoustic Siren & Brevo Emergency Broadcast]